Algebraic Formulation of the $3n + 5$ Collatz Orbit in Jacobsthal Space: The $3J + 5J(0,1)$ Transformation

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Table 1. Collatz Orbit for Initial Node 347 in J-Spinor Representation under $3n+5$ Dynamics

Step	Odd Node	Standard J	Reduced J	3J + 5J(0,1)	Next J
0	347	J(176, 347)	J(176, 347)	J(528, 1046)	J(264, 523)
1	523	J(264, 523)	J(264, 523)	J(792, 1574)	J(396, 787)
2	787	J(396, 787)	J(396, 787)	J(1188, 2366)	J(594, 1183)
3	1183	J(594, 1183)	J(594, 1183)	J(1782, 3554)	J(891, 1777)
4	1777	J(891, 1777)	J(224, 443)	J(672, 1334)	J(336, 667)
5	667	J(336, 667)	J(336, 667)	J(1008, 2006)	J(504, 1003)
6	1003	J(504, 1003)	J(504, 1003)	J(1512, 3014)	J(756, 1507)
7	1507	J(756, 1507)	J(756, 1507)	J(2268, 4526)	J(1134, 2263)
8	2263	J(1134, 2263)	J(1134, 2263)	J(3402, 6794)	J(1701, 3397)
9	3397	J(1701, 3397)	J(848, 1701)	J(2544, 5108)	J(1277, 2549)
10	2549	J(1277, 2549)	J(636, 1277)	J(1908, 3836)	J(959, 1913)
11	1913	J(959, 1913)	J(118, 241)	J(354, 728)	J(182, 359)
12	359	J(182, 359)	J(182, 359)	J(546, 1082)	J(273, 541)
13	541	J(273, 541)	J(134, 273)	J(402, 824)	J(206, 407)
14	407	J(206, 407)	J(206, 407)	J(618, 1226)	J(309, 613)
15	613	J(309, 613)	J(152, 309)	J(456, 932)	J(233, 461)
16	461	J(233, 461)	J(114, 233)	J(342, 704)	J(176, 347)
17	347	J(176, 347)	-	-	LOOP DETECTED

The J-spinor transition process of the Collatz orbit from the initial node $b_0 = 347$ to its loop closure (node 347). Each column represents the following algebraic operations:

- **Odd Node:** The current odd node b in the standard $3n + 5$ Collatz orbit.
- **Standard J:** The standard spinor $J(a, b)$ defined by the internal state $a = (b + 5)/2$.
- **Reduced J:** The reduced spinor extracted via J-space 2-adic reduction, serving as the foundational input vector for the current step.
- **3J + 5J(0,1):** The result of applying the affine transformation to the reduced input spinor.
- **Next J:** The normalized standard spinor of the next odd node, obtained through sequence expansion and normalization without introducing fractional components.

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